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Q1. A process runs in a system with multi level paging and it has a logical address space of 8 bits. In the system page size is 16 Bytes, size of each entry of the page table is 4 Bytes and size of the main memory is 512 Bytes. In order to fit the pages of the process in the main memory the OS applies a two-level paging technique in outer page number bits of the logical address space until the outer most page table can be allocated in a frame of the main memory. **[3+3]**

- I. Illustrate the logical address space of the process including the necessary outer page bits, inner page bits and offset bits of every step with proper mathematical calculations during the paging mechanism of the system described above.
- II. In this system, if the CPU generates logical addresses **90** then map the corresponding physical addresses of these logical addresses.
Necessary page table information is given below:

Outer page table		
Page #	Frame #	valid/invalid bit
0		i
1	6	v
2	11	v
3		i

Inner page tables					
PMT at frame 6			PMT at frame 11		
Page #	Frame #	valid/invalid bit	Page #	Frame #	valid/invalid bit
0	3	v	0	25	v
1	20	v	1		i
2		i	2	12	v
3	7	v	3	14	v

Q2. As an upcoming OS enthusiast, You decided to build a Page Table System. The Page size you settled on is **4KB(2^{12} Bytes)** per page. You also determined that each page table entry will take **64 Bytes(2^6 Bytes)**. You are making a **two level paging system**. Now with this info, what will be the **maximum size of a process** that you can manage in this system? **[4]**